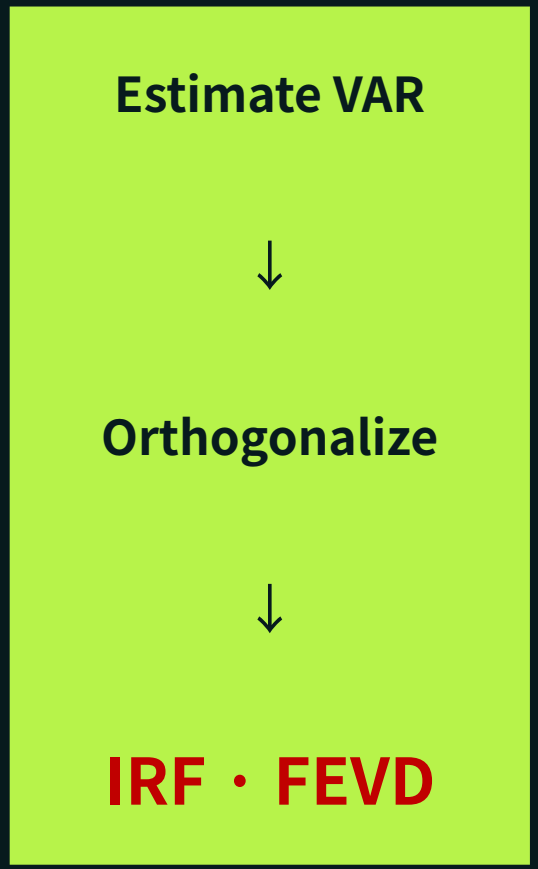
Impulse Responses and Variance Decomposition

Two tools for reading a VAR — IRF and FEVD



Financial Time Series · 05

This section moves from questions to responses, shares and cases

01	Background	**Why VAR coefficients are not enough** Coefficient count · indirect paths · 3 questions
02	Impulse responses	**IRF — size, sign, persistence** VMA form · orthogonalization · ordering
03	Variance decomposition	**FEVD — relative importance** Forecast error split · share formula · ordering
04	Interpretation and cases	**Reporting checklist and four cases** Monetary policy · oil · spillover · stress test

Two summaries of one VAR — the IRF shows the path, the FEVD shows the share

VAR coefficients alone do not reveal how shocks propagate

A VAR(1) without intercept — the variables move on their own past and new shocks

Basic VAR(1)

$$z_t = A\, z_{t-1} + e_t$$

Coefficient count

Coefficients pile up

2 variables, 4 lags: 16 slopes, mixed signs

Indirect paths

Shocks loop back

Single coefficients miss the cumulative path

Limit of Granger causality

Only yes or no

Silent on how much, how long, how important

So instead of reading coefficients, we feed in a shock and summarize the outcome

Granger asks whether, the IRF asks how, the FEVD asks how much

All three come from the same VAR but ask different questions and produce different outputs

Granger causality
Predictive content

Question

Does the past of x help forecast y?

Yes / No

Output

Yes / No (F-test)

IRF
Impulse response

Question

How much does y move h periods after an x shock?



Output

Response path + bands

FEVD
Variance decomposition

Question

What % of y's forecast uncertainty is due to x?



Output

Share (%) by horizon

※ Granger causality = F-test (yes/no); IRF = response curve by horizon; FEVD = table of shares (%) by horizon

Today's value is a weighted sum of past shocks

Substituting the VAR(1) into itself repeatedly gives the VMA (vector moving average) form

VMA form and response matrices

$$z_t = e_t + A\,e_{t-1} + A^2 e_{t-2} + A^3 e_{t-3} + \cdots, \qquad R_h = A^h$$

Response matrix

Element (i, j) = effect

Effect of shock j on variable i after h periods

First term

At h=0 it is I

A new shock enters only its own variable, by 1

Stability condition

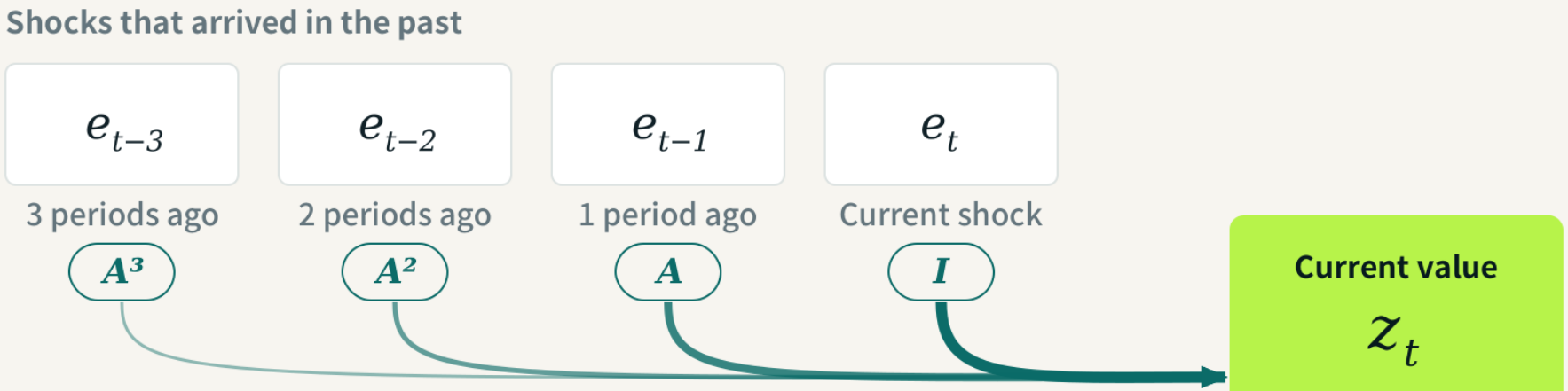
All |eigenvalues| < 1

A to the h goes to 0, so shocks die out

An IRF is simply these weights plotted against the horizon

Each multiplication by A shrinks the shock as it spreads

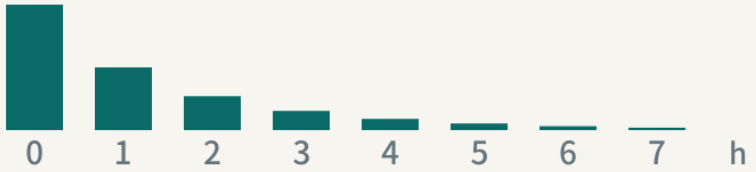
Note example VAR(1) — the eigenvalues of A are 0.6 and 0.3



Weight size (A to the h, y→y element)

Eigenvalues 0.6 and 0.3 are below 1

→ shocks die out over time

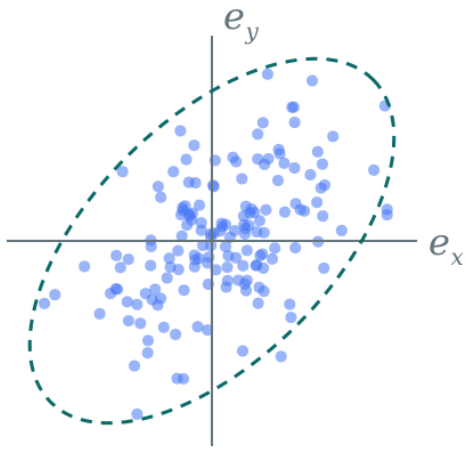


※ Bars = (y, y) element of A to the h (1, 0.5, 0.27, …). A VAR(p) builds the same weights recursively

Reduced-form shocks are correlated, so they must be orthogonalized

If the shock covariance matrix S is not diagonal, one shock cannot be moved alone

Problem — reduced-form shocks co-move

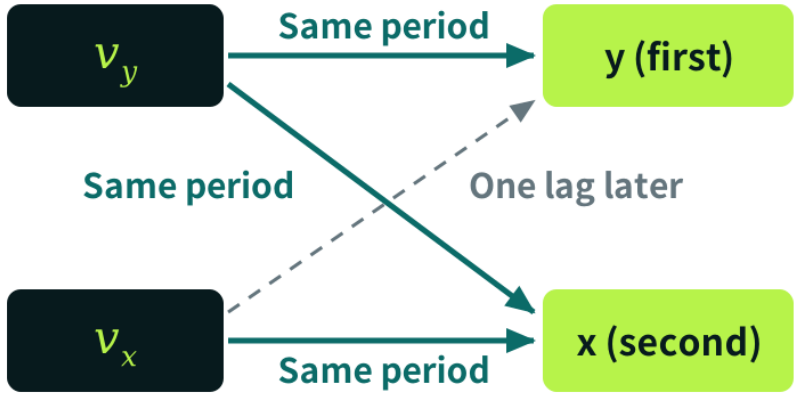


Corr. 0.5

When an x shock hits, the y shock moves with it

An "x shock only" experiment fails

Solution — orthogonalize with ordering [y, x]



Only shocks to earlier variables hit later ones at once
→ **the ordering itself is an economic assumption**

※ **Convention: slow-moving variables (output, prices) first; fast-moving ones (interest rates, stock prices) last**

Cholesky factorization yields uncorrelated structural shocks

Factor S into a lower-triangular P and the structural shocks get an identity covariance

Cholesky factor and structural shocks

$$S = PP', \qquad v_t = P^{-1}e_t, \qquad \mathrm{Cov}(v_t) = I$$

Orthogonalized IRF

$$O_h = R_h P$$

Element = 1 s.d. response

Response of variable i, h periods after a one-s.d. structural shock j

Recursive assumption

Earlier shocks hit later variables at once; the reverse takes a period

Once P multiplies the responses, the variable ordering enters the results

The note example assigns the 0.5 shock correlation to y

VAR(1), ordering [y, x], both shock variances 1, covariance 0.5

Example coefficients and Cholesky factor

$$A = \begin{pmatrix} 0.5 & 0.2 \\ 0.1 & 0.4 \end{pmatrix}, \quad S = \begin{pmatrix} 1 & 0.5 \\ 0.5 & 1 \end{pmatrix} \Rightarrow P = \begin{pmatrix} 1 & 0 \\ 0.5 & 0.866 \end{pmatrix}$$

Response at h=0 = P

Because the first response matrix is I

x shock → y: 0

The ordering with y first shows up directly

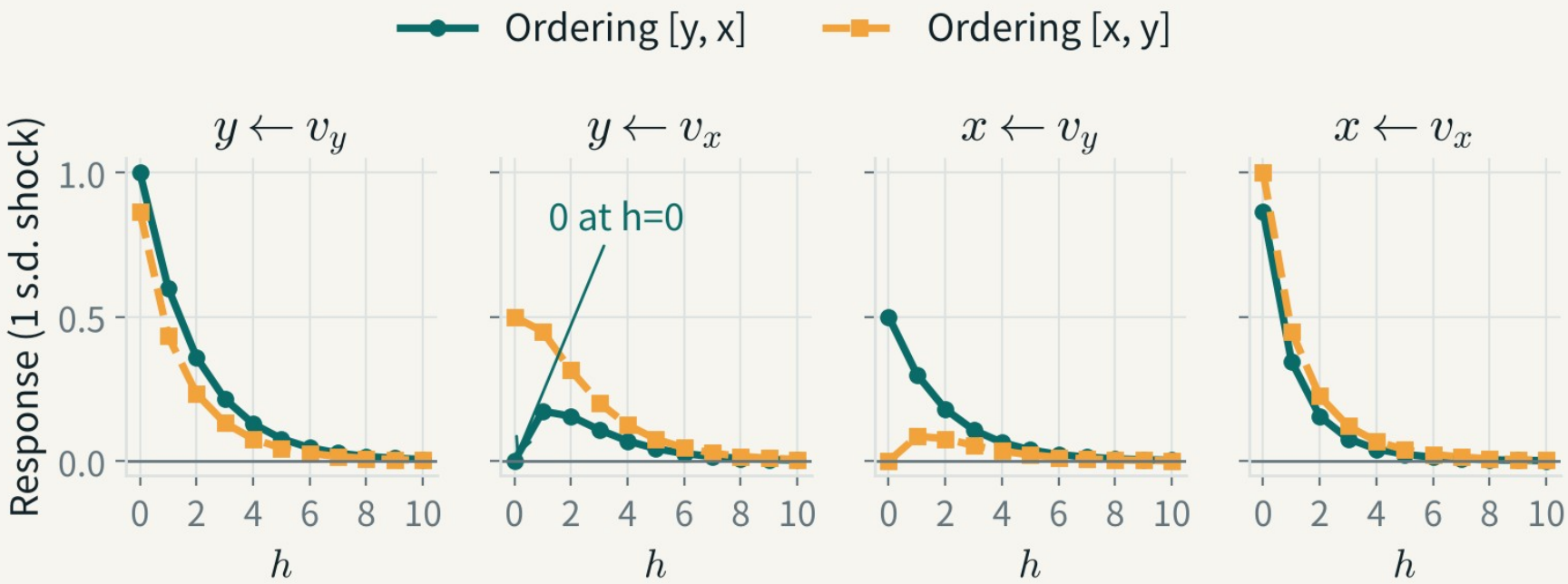
y shock → x: 0.5

The 0.5 correlation is assigned from y to x

With the same covariance, who comes first decides the h=0 response

Reversing the ordering changes the IRFs of the same VAR

Orthogonalized responses up to h=10 — solid: ordering [y, x]; dashed: ordering [x, y]



※ numpy, note example (A, S). Ordering [y, x], h=0–2 matches the note table (1.000 · 0.600 · 0.360 …)

Other IRFs avoid or complement the ordering dependence

Generalized IRF · structural VAR · cumulative IRF and local projections

Pesaran & Shin (1998)

Generalized IRF

- Ordering-invariant
- Keeps correlations
- Not structural

Theory-based identification

Structural VAR

- Zero restrictions
- Blanchard–Quah
- Sign restrictions

Differenced data · checks

Cumulative IRF · LP

- Sum = level effect
- LP: direct per h
- LP: noisier

※ Bands come from bootstrap or Monte Carlo. A band that covers 0 means no significant response

A variance decomposition built from GIRFs does not sum to 100%

Forecast error variance splits into sums of squared responses by shock

Structural shocks are uncorrelated with unit variance, so the squared sums separate cleanly

h-step forecast error = effect of the structural shocks arriving over the next h periods

$$z_{t+h} - \hat{z}_{t+h|t} = O_0\, v_{t+h} + O_1\, v_{t+h-1} + \cdots + O_{h-1}\, v_{t+1}$$

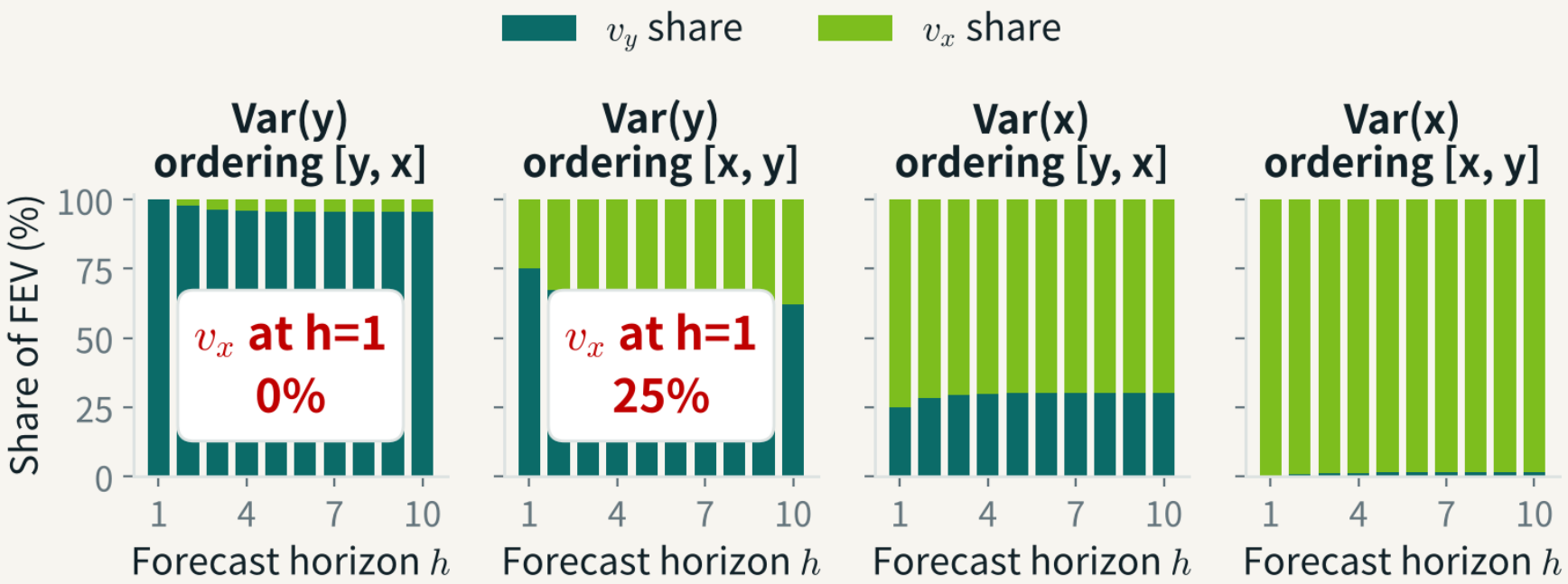
Share of shock j in the forecast error variance of variable i

$$\mathrm{share}_{ij}(h) = \frac{\sum_{s=0}^{h-1} o_{ij,s}^2}{\sum_{k=1}^{N} \sum_{s=0}^{h-1} o_{ik,s}^2}, \qquad \sum_{j=1}^{N} \mathrm{share}_{ij}(h) = 1$$

The denominator is the forecast error variance (MSE); all shares sum to 100%

The ordering alone moves the x-shock share of Var(y) from 0% to 25%

FEVD of the note example up to h=10 — same data, same VAR



※ Ordering [y, x], h=1·2·3: x in Var(y) 0 → 2.2 → 3.5%; y in Var(x) 25.0 → 28.1 → 29.4% (as in note)

How the own-shock share evolves over horizons reveals a variable's role

A rule of thumb for FEVD — look at how shares change with the horizon, not just their level

Own share stays high

Exogenous variable

- Rarely pulled
- E.g. y in [y, x]
- Long run ≈ 95%

Others' share grows

Driven variable

- Effects accumulate
- E.g. y in Var(x)
- 25% → about 30%

Reporting principle

Check robustness

- Reorder, recompute
- Argue from theory
- State the horizons

The FEVD also depends on the orthogonalization, that is, on the ordering

Reporting IRFs and FEVDs involves seven checks

Check in order: shock definition → presentation of results → model diagnostics

Shock definition

- ☑ **Shock size**
  One s.d. or one unit?
- ☑ **Ordering**
  Rationale, plus results under other orderings

Presentation

- ☑ **Differenced data**
  Add cumulative IRF for level effects
- ☑ **Confidence bands**
  Band covers 0 = no significant response
- ☑ **FEVD horizons**
  State them, e.g. 1, 4, 12 periods

Model checks

- ☑ **Stability**
  If unstable, the IRF never dies out
- ☑ **VECM**
  May converge to a permanent effect

※ **Orthogonalized IRFs in statsmodels are scaled to a one-standard-deviation shock**

Monetary policy IRFs change shape with the variable set

Case 1 · Christiano, Eichenbaum & Evans (1999) — orthogonalized IRFs

Setup

Policy rate last

- Output & CPI first
- Rate reacts fast
- 1 s.d. tightening

Result

Output falls gradually

- Slow, steady drop
- Prices rise first
- = price puzzle

Lesson

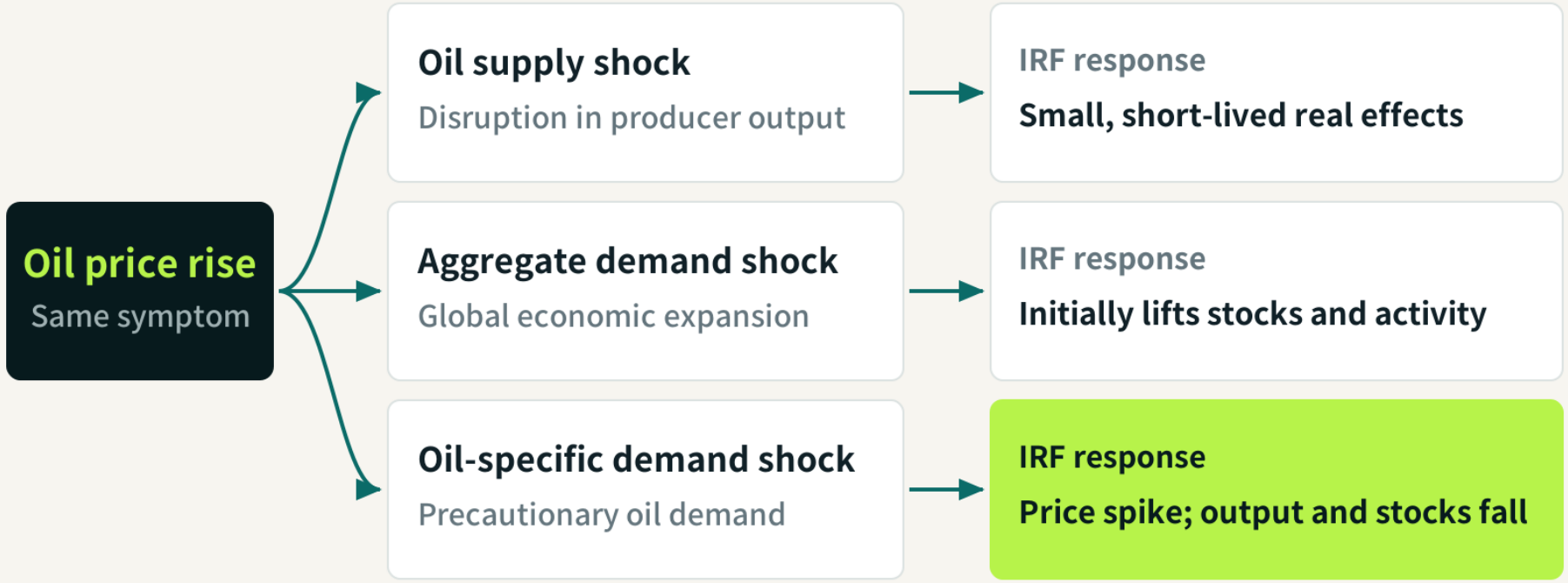
Add commodities

- Puzzle eases
- Set shapes the IRF
- Report the scheme

The name of a structural shock comes from the identifying assumptions

Why oil prices rose determines how asset prices respond

Case 2 · Kilian (2009), Kilian & Park (2009) — oil shocks decomposed by cause with structural VAR IRFs



※ One oil price rise, three causes: real and stock responses differ across supply, demand and precautionary shocks

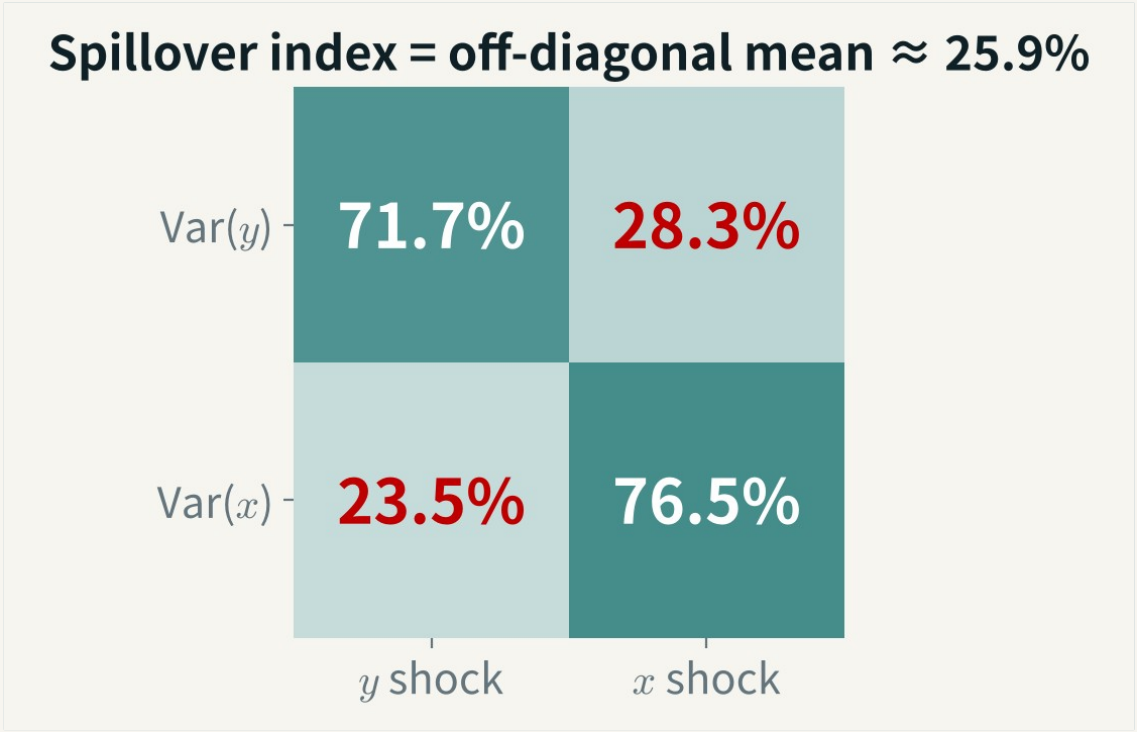
Pooled off-diagonal FEVD shares form a spillover index

Case 3 · Diebold & Yilmaz (2009, 2012) — connectedness based on the generalized FEVD

Off-diagonal = others' shocks

- GFEVD avoids ordering
- Rows normalized to 100%
- Rolling: jumps in crises

Watch when it spikes



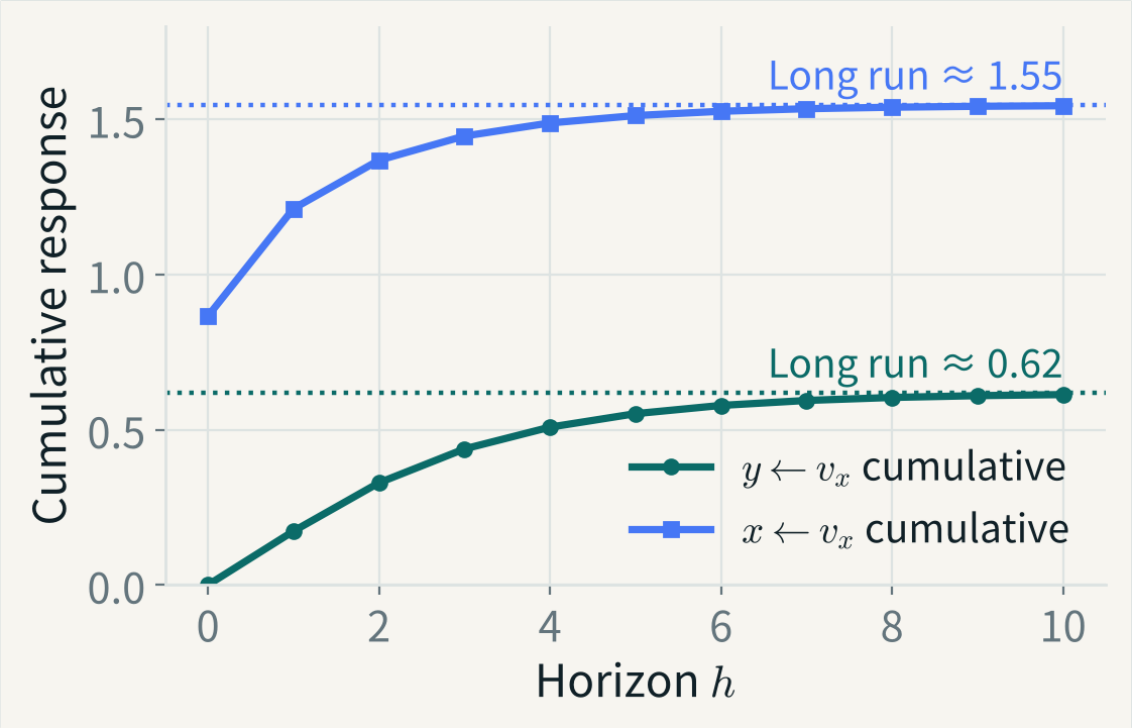
Cumulative IRFs add a propagation path to stress scenarios

Case 4 · pension fund stress test — cumulative response to the x shock in the note example

Rate and FX shocks

- Response by asset class
- Portfolio P&L path
- Show adverse bands too

VaR misses the time path



Lab 1 checks monetary policy IRFs and ordering robustness in one pass

Claude Code prompt — FRED monthly data, 1965-01 to 2007-12

Data

INDPRO · CPIAUCSL · PPIACO · FEDFUNDS

Logs except the rate; ordering [output, prices, commodities, rate]

Estimate the VAR → **Rate shock orthog. IRF** → **FEVD table (1, 2, 4 yrs)** → **Rate first: recompute** → **Save .md, open figures**

※ Students paste the prompt and check results — Claude Code handles downloads, runs and error fixes

Same data, only the ordering changes — how much do the IRF and FEVD move?

Lab 2 computes a spillover index for global equity markets

Claude Code prompt — five yfinance indices, weekly log returns since 2005

Data

US · Europe · Japan · Korea · Hong Kong

Daily closes of ^GSPC · ^STOXX50E · ^N225 · ^KS11 · ^HSI

Weekly returns, align dates → **Pick lags by BIC, fit VAR** → **Implement 10-week GFEVD** → **Heatmap, 100-wk rolling** → **Korea: sender or receiver?**

※ If statsmodels lacks GFEVD, have Claude implement the formula. Mark 2008, 2020 and 2022 on the chart

Spikes in the spillover index are a systemic risk signal

The IRF shows how a shock spreads;

the FEVD shows each shock's share of uncertainty.

Ordering is an assumption — vary it and report it.

Next: in the workbook deck,
compute Cholesky and FEVD by hand